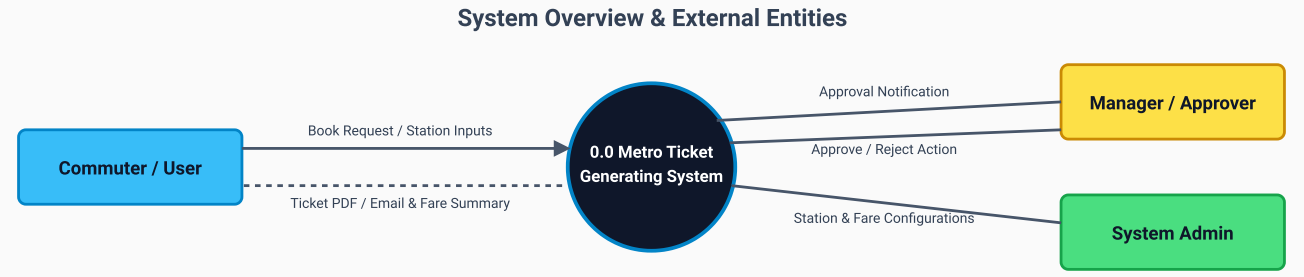
Project Design Phase: Data Flow Diagrams

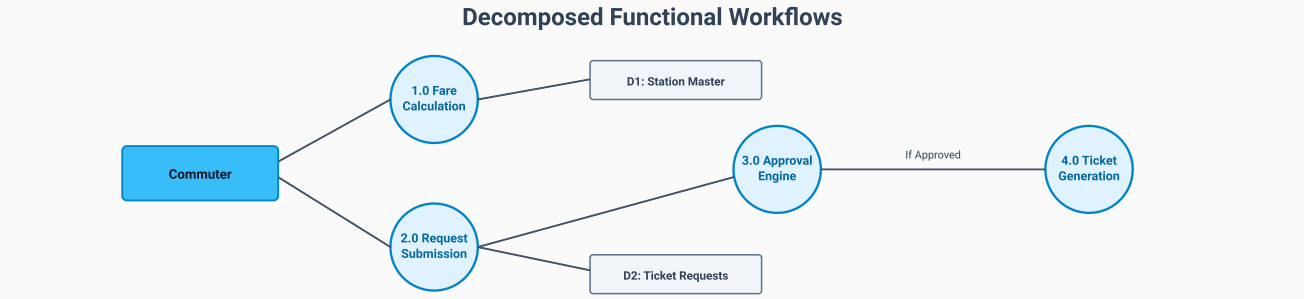
METRO TICKET GENERATING SYSTEM IN SERVICENOW

DOCUMENT TYPE	PHASE	PLATFORM	STATUS
DFD & User Stories	3. Project Design Phase	ServiceNow Portal	Approved Blueprint

1. CONTEXT LEVEL DIAGRAM (DFD LEVEL 0)



2. SUB-SYSTEM PROCESS FLOW (DFD LEVEL 1)



3. FUNCTIONAL USER STORIES & BACKLOG MATRIX

Below is the agile backlog specification detailing requirements, user roles, acceptance criteria, and priority rankings.

ID	USER TYPE	USER STORY (REQUIREMENT)	ACCEPTANCE CRITERIA	PRIORITY	RELEASE
USN-01	Commuter	As a user, I want to select source and destination stations on the portal form.	Dropdown auto-populates valid active metro stations dynamically.	HIGH	Sprint 1
USN-02	Commuter	As a user, I want to see real-time fare calculation before submitting.	Fare updates instantly upon selecting stations and pass type.	HIGH	Sprint 1
USN-03	Commuter	As a user, I want form validation preventing identical source and destination.	System displays inline warning and disables submit button.	MEDIUM	Sprint 1
USN-04	Approver	As a manager, I want automated email notifications for pending ticket approvals.	Email arrives with direct "Approve" and "Reject" action links.	HIGH	Sprint 2
USN-05	Commuter	As a user, I want an automated email with my ticket upon approval.	Email contains unique ticket number, route details, and QR code.	HIGH	Sprint 2
USN-06	Commuter	As a user, I want to view my past ticket request status on the Service Portal.	Portal status shows Pending, Approved, Generated, or Cancelled.	MEDIUM	Sprint 2
USN-07	Commuter	As a user, I want to cancel my ticket and initiate refund processing.	Cancellation updates status to "Cancelled" and logs refund amount.	MEDIUM	Sprint 3
USN-08	System Admin	As an admin, I want to configure master fare matrices and station routes.	Admin table view allows updating station lists and fare rules.	HIGH	Sprint 1
USN-09	Developer	As a developer, I want Flow Designer automation to manage workflow states.	Flow automatically advances record state based on approval outcomes.	HIGH	Sprint 1
USN-10	System Admin	As an admin, I want scheduled jobs to perform automated nightly fare audits.	Scheduled script verifies unused tickets and clears expired states.	LOW	Sprint 3